

- 1 (a) With reference to Fig. 1.1, by means of a labelled arrow, indicate on the diagram the most suitable position to insert the fragment of human insulin cDNA.

anywhere within the lac Z gene (i.e. map position 146 – 469);

[1]

- (b) Using the information given in Table 1.1, select a restriction enzyme that will generate a blunt end and also allow for subsequent identification of successful recombinant colonies.

[1]

(i) State your choice of restriction enzyme: SmaI;

(ii) Explain your choice of restriction enzyme.

[1]

disrupt lacZ gene will allow visual confirmation/ see white colonies (OWTTE);
reject 'selectable marker' (since antibiotic/ bla gene/ amp^R can also be one)

- (c) Some restriction enzymes generate sticky ends, while others generate blunt ends. Explain why it is more efficient to clone DNA fragments with sticky ends than DNA fragments with blunt ends.

temporary hydrogen bonds to hold 2 fragments (via overhangs) in place while DNA ligase seals nicks;

[1]

- (d) Describe how a set of blunt ended DNA fragment could still be efficiently cloned into a plasmid vector.

mentions → polylinker/ linker/ adaptor;

aware → that the artificial sequence is double-stranded;

describes → see below for both options

If polylinker/linker employed:

cDNA with blunt fragment and vector are cleaved with same restriction enzyme/ or answer in B)i) that generated blunt ends;

Ligate polylinker(s) to both cDNA fragment and vector;

Linker contains restriction site that can produce fragments with sticky end;

Cut/ cleave with same restriction enzymes to generate complementary sticky/overhangs;

Add DNA ligase into tube containing both fragments and allow them to ligate and form recombinant DNA molecule;

If adaptor employed:

cDNA fragment and vector need not be cleaved with same restriction enzyme;

Because adaptors have one sticky end and the other is a blunt end (that joins to the cDNA molecule/ foreign gene);

Select appropriate adaptor with complementary overhang/ end to the vector molecule;

[3]

- (e) Outline the principle of insertional inactivation of the lacZ gene.

method allows visual confirmation of recombinant colonies;

culture medium contains X-gal, a chromogenic substrate that when cleaved by β -galactosidase, will turn blue;

	Non recombinant	Recombinant
State colour;	Blue	White
Explain	Complementation of lacZ' gene (in plasmid) and lacZ' gene in <i>E. coli</i> ; hence functional β -galactosidase expressed	gene of interest inserted, disrupts lacZ' gene (in plasmid); hence functional β -galactosidase not expressed

[4]

total of 5 marking points

After transforming the clones into the *E. coli* cells, the cells were plated onto appropriate mediums. **Positive results were observed** from the morphologies of the grown colonies. The colonies were harvested and purified **via affinity chromatography**. A portion of the purified extract was further analysed. Results of the analysis revealed that **functional insulin** was not present in the extract.

- (f) Suggest two reasons why functional insulin was not expressed in the recombinant *E. coli* cells.

2 ways of explaining →

- 1) inserted in wrong direction/ orientation;
resulting in transcription of incorrect strand, hence encoding for a different polypeptide/ or correct/ functional polypeptide is not produced;
- 2) the α -chain and β -chain of insulin were cloned in separate vectors;
since gene of interest was inserted next to lacZ gene, therefore fusion polypeptide/ insulin- β -gala polypep was expressed;
functional insulin can only be achieved when the fusion polypeptide is cleaved by CBr/ two separate chains are joined by disulfide bridges;

[4]

total of 5 marking points

[Total: 15 marks]

- 2 (a) With reference to Fig. 2.2, explain whether the fetus would be affected with cystic fibrosis?

yes, fetus is affected with cystic fibrosis;
due to **deletion of three base pairs** encoding for the phenylalanine;
mutation results in shorter allele/ fragment seen in Fig 2.2 compared to normal allele;
parents are heterozygotes, carrying 1 normal allele and 1 diseased allele;
fetus is homozygote, carrying 2 copies of the diseased allele;
both alleles located on the homologous chromosome;

[5]

- (b) Explain how Tsp45I cuts at the indicated site.

recognize specific **DNA** sequences;
shape of DNA sequence/ DNA sequence is complementary to **active site** of enzyme;

[2]

- (c) A primer has been synthesized to amplify a 200bp segment of exon 4 containing the Tsp45I site, and that the Tsp45I site is 80bp from the right primer. With reference to PCR and gel electrophoresis, describe how α -synuclein mutation is detected in a Parkinsonian patient.

PCR-RFLP method;
isolate genomic DNA from the Parkinsonian patient;
PCR to amplify the 200-bp segment of exon 4;
purify the PCR product and digest it with Tsp54I;
separate the products by size using gel electrophoresis;
normal allele will contain the Tsp45I site, and so produce 2 fragments/120- and 80-bp fragments;
the mutant allele will not contain the Tsp45I site and so produce one fragment/ only a 200-bp fragment;

[5]

- (d) Suggest the sizes of fragments that could be obtained from homozygotes for the normal allele, homozygotes for the mutant allele and in heterozygotes?

homozygotes for the normal allele will have 120- and 80-bp fragments;
homozygotes for the mutant allele will have 200-bp fragments;
heterozygotes will have 200- and, 120- and 80-bp fragments;

[3]

[Total 15m]

3 (a) Explain what is meant by a transgenic variety of papaya?

exogenous/foreign gene;
integrated into its genome;
through recombinant DNA techniques;
and expressed;
foreign gene is from virus;

[2]

(b) Suggest why farmers in the Hawaii grow papaya carrying the PRSV gene?

pathogen derived resistance;
transforming plants with pathogen's gene would result in the transgenic plant being resistant;
gene from the pathogen was being used to fight against the pathogen itself;

[2]

(c) Apart from using *Agrobacterium*, suggest one method by which the PRSV gene can be introduced into the plant.

electroporation/ microinjection (microprojectile bombardment) / gene gun;

[1]

(d) With reference to Fig. 3.1,

(i) explain what is meant by *haploid callus*.

haploid → 1 set of chromosome/ half the diploid no. of chromosomes (reject only 'n')
callus → undifferentiated mass of cells derived from protoplasts/single plant cells;

[2]

(ii) state **one** precaution that should be taken when transferring material onto culture medium **A** or **B**.

sterile conditions; reject – controlled environmental conditions

[1]

(iii) state **one** similarity and **one** difference in the composition of the media between culture medium **A** and culture medium **B**.

similarity presence of nutrients;
difference **A** induces mitosis;
B induces differentiation;

Named correct/ relevant plant growth substance(s);

[3]

(iv) explain why the final plants produced are homozygous at all loci.

spontaneous doubling/ existing chromosomes were doubled up;
via sister chromatids remaining in the same cell;
due to spindle failure at meiosis;
artificially induced by colchicine;

Comments:

- candidates tended not to know what colchicine actually does in mitosis

[2]

(e) Explain the advantages of tissue culture over more traditional methods of cloning plants, such as taking cuttings or graftings.

much larger numbers can be produced/ mass production;
requires less space;
(cultures) easy to transport/ ref. air freight;
ensures new plants are disease-free;
genetic modification possible;
not limited by seasons/ external conditions/ weather;

[2]

[Total: 15m]

- 4 (a) The ability of complementary nucleotides to base pair using hydrogen bonding, and the ability to selectively disrupt or retain accurate base pairing by treatment with chemicals and/or heat is critical to many methods used to produce and analyze recombinant DNA. Give 3 examples of methods that rely on complementary base pairing, and explain what role complementary base pairing plays in each of these methods.

The 3 examples:

1. ligation of foreign DNA fragment with sticky ends with complementary sticky ends in a cloning/plasmid vector;
2. annealing of a labeled nucleic acid probe with a single stranded DNA or RNA molecule attached to a membrane during library screenings/Southern hybridization;
3. annealing of primer to single stranded (DNA or RNA) template during PCR;

Roles:

(General) Base pairing allows for nucleotides to interact in a sequence-specific manner essential for the procedure's success;

1. formation/ creation of recombinant DNA molecule;
2. detection of membrane-bound single-stranded nucleic acid sequences in hybridization of probes to Southern blots requires complementary base pairing between a bound strand of DNA and complementary sequences in the probe;
3. extension/ elongation of complementary strand – Taq DNA polymerase adds nucleotides to the 3' end of the existing primer to synthesize a complementary strand;

[6]

- (b) Outline how *E. coli* can be genetically engineered to produce insulin avoiding the problems associated with introns.

1. Construction of recombinant DNA molecule (foreign + vector DNA)
 - a. Source of gene - mRNA for insulin
 - Isolate mRNA **from** pancreas/ beta cells of pancreatic cells;
 - **RT** is used to reverse transcribed mRNA to form complementary DNA (cDNA);
 - **DNA pol** then synthesizes the second DNA strand;
 - Mentioned use of adaptors/ linkers to create sticky ends (since cDNA will be blunt);
 - b. Creation of recombinant DNA molecule
 - Vector and insulin gene must be cleaved with same restriction enzyme at specific recognition sequence/site;
So that complementary base pairing can form via hydrogen bonds between insulin gene fragment and vector;
 - Example (of vector/ restriction enzyme);
 - DNA ligase is added to seal phosphodiester bond;
2. Introduction of recombinant DNA molecule into living host (e.g. bacteria)
 - Via transformation;
 - Method of transformation;
 - Reason of method (to create permeability/ transient enlargement of pores);
 - Purpose of living host (provide enzyme machinery);
3. Multiplication of recombinant DNA molecule and cell growth/ division;
4. Screening/ Selection & identification of clones with recombinant DNA molecules [4 marks max]

Insertional inactivation;
Lac Z/ Blue-white selection or antibiotic selection;
Description of either lac Z or antibiotic selection (max 2 marks);
5. Purification of insulin product

Upscale/ production (max 1 mark);
Post-translational modification/ combining the 2 chains via disulphide bonds; [max 1 mark]

[8]

(c) Nov 98 Op 3 Qn 3a i) / ii)
Explain the principles of genetic fingerprinting.

- principle/ based on DNA polymorphism;
- non-coding tandemly repeated DNA sequences/VNTRs/ SNPs/ STRs/ SSRs/ microsatellites exist;
- sequences are inherited, half of an individual's VNTRs derived from the mother and half from the father (being inherited in the same way as different alleles of coding genes);
No person has VNTRs that are identical to those of either parent (this can only happen as a result of cloning)
- Each variant acts as an inherited allele (hence loci/locus), allowing them to be used for identification/ act as scientific marker/ DNA fingerprint;
- different number/ molecular weight/ length of restriction fragments generated among individuals when cut/ cleaved with the same restriction enzyme at a given sequence
- run sample on gel electrophoresis/ nucleic acid hybridization/ Southern blotting to separate DNA fragments;
- identification/ analysed/ compared against another individual/ known sample to detect matching band(s)/pattern
- Because these repeated regions can occur in many locations, the probes used to identify them complement the DNA regions that surround the specific microsatellite being analyzed.
- Application areas → e.g. disease detection/ paternity testing/ forensic analysis for crime investigation/ trace evolutionary lineage/ genomic mapping; [6]

[Total : 20]